

Research Statement

Introduction

My research lies at the intersection of stable homotopy theory and arithmetic, with a particular emphasis on equivariant phenomena. The central goal of my research program is to understand two related aspects of *Real algebraic K-theory*, its duality phenomena and its descent behavior. Duality concerns structural relationships among its invariants, while descent asks how global information can be recovered from local or cohomological data while preserving equivariant structure.

My dissertation established the *2-adic K-theoretic Tate–Poitou duality* for number fields, providing the non-equivariant starting point in this program. I am now extending this duality picture to Grothendieck–Witt theory, the fixed-point side of Real algebraic *K*-theory, toward a C_2 -equivariant duality. In parallel, my work on *Real trace methods* develops computational tools for Real algebraic K-theory.

Developing such a descent theory leads to foundational questions in equivariant algebra. A genuine equivariant form of descent must have operations and representation-graded information that disappear after forgetting the group action. My work on *transfer systems* and $\mathcal{RO}(C_2)$ -gradings studies these two kinds of structure algebraically. In the longer term, I aim to develop genuine equivariant forms of cohomology and descent and to understand the arithmetic duality they should carry.

1 Background

A simple example of duality comes from homology, which studies subspaces, and cohomology, which studies functions on those subspaces. The relationship resembles that between land and trees, with the land determining what can grow and the trees in turn revealing the nature of the land beneath them. Mathematically, the *Universal Coefficient Theorem* makes this relationship precise. For example, over a field k , it identifies

$$H^n(X; k) \cong \text{Hom}_k(H_n(X; k), k).$$

A related principle is *descent*, which reconstructs global information from compatible local or cohomological data.

These principles become subtler when the objects under study carry symmetries. Two actions can have the same underlying object after the symmetry is forgotten while having very different equivariant information. For instance, reflection and rotation by a half-turn give two C_2 -actions on a circle with different fixed-point sets, even though both have the same underlying circle. My work asks how duality and descent can be

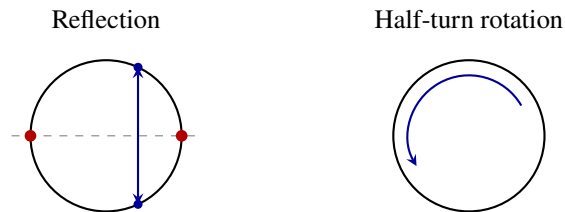


Figure 1: Reflection fixes two red points, while rotation by half a turn fixes none.

formulated without discarding the symmetry.

Real algebraic *K*-theory (KR) provides a natural setting for this question. Ordinary algebraic *K*-theory studies projective modules over rings, while KR-theory incorporates an involution and records the resulting C_2 -equivariant structure. Forgetting this symmetry recovers K-theory, while taking fixed points leads to

Grothendieck–Witt theory (GW). Thus, a duality in KR-theory should be visible both in K-theory GW-theory, together with compatibility between them.

Arithmetic gives a concrete setting in which to study this interaction. In arithmetic, Tate–Poitou duality is expressed through étale cohomology, while descent connects this cohomological information to algebraic K -theory. Arithmetic duality should therefore also appear at the level of K -theory, and its Real refinement should preserve both the duality and the equivariant structure underlying descent.

2 Arithmetic Duality and Real Algebraic K-theory

Real algebraic K -theory brings ordinary algebraic K -theory and Grothendieck–Witt theory into a single C_2 -equivariant framework. My dissertation established an arithmetic duality on the non-equivariant side through K -theoretic Tate–Poitou duality at the prime 2. My current work develops the corresponding fixed-point counterpart and shows that this duality extends across the hyperbolic sequence $U \rightarrow K \rightarrow GW$. Together, these results suggest a larger equivariant duality in Real algebraic K -theory.

2.1 K-theoretic Tate–Poitou duality. Tate–Poitou duality is expressed through étale cohomology, while Thomason descent relates this cohomological information to $K(1)$ -local algebraic K -theory. The corresponding K -theoretic problem is to realize Tate–Poitou duality at homotopical level.

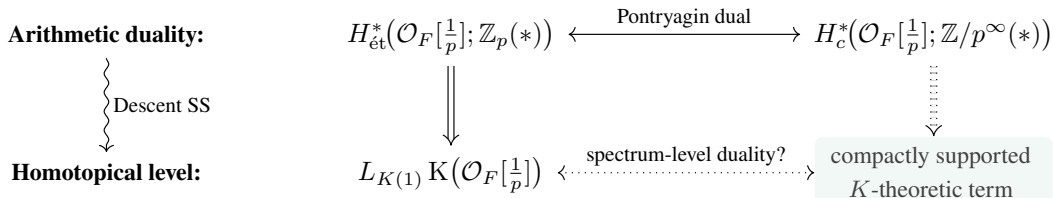
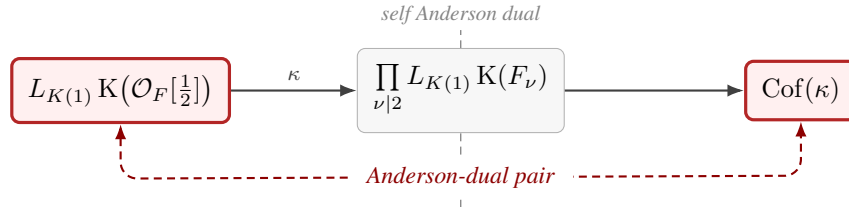


Figure 2: Cohomological duality suggests a spectrum-level duality after Thomason descent.

A homotopy-group-level form of this duality was conjectured by Calegari in [CE12], and Blumberg and Mandell established the corresponding spectrum-level duality at every odd prime in [BM20]. The prime 2 remained open because the presence of real embeddings introduces additional cohomological behavior. My dissertation resolves this remaining case.

Theorem 1 ([Cho25]). *Let F be a number field and $\mathcal{O}_F[\frac{1}{2}]$ be its ring of 2-integers. The completion cofiber sequence of $K(1)$ -local algebraic K -theory admits a canonical 2-adic Anderson self-duality. The duality acts by reflection, interchanging the end terms and identifying the middle term with its own dual.*



The theorem has a concrete application to trace methods in the algebraic K -theory of the sphere. For $F = \mathbb{Q}$, combining the duality theorem with known comparison results gives the following identification.

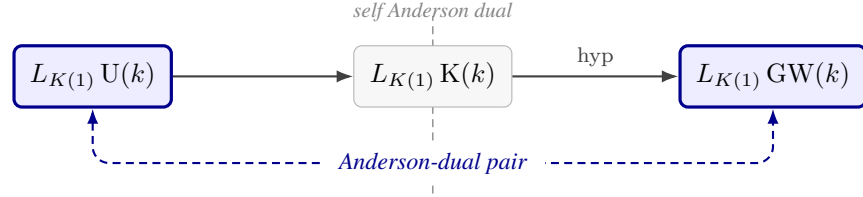
Corollary 2. *There is a canonical equivalence in nonnegative degrees*

$$\Sigma^{-1} I_{\mathbb{Z}_2} L_{K(1)} K(\mathbb{Z}) \simeq_{[0, \infty)} \text{Fib} \left(K(\mathbb{S})_2^\wedge \xrightarrow{\text{trcs}} \text{TC}(\mathbb{S})_2^\wedge \right).$$

Thus, arithmetic information in $K(\mathbb{Z})$ determines the fiber of the cyclotomic trace map for $K(\mathbb{S})$.

2.2 Grothendieck–Witt theory and duality. A fixed-point counterpart of the K-theoretic duality naturally leads to Grothendieck–Witt theory, which records projective modules equipped with nondegenerate bilinear forms. The hyperbolic map $K \rightarrow GW$ relates algebraic K-theory to this additional form structure, whose homotopy fiber is known as U-theory. The 2-primary descent techniques from my dissertation provide the arithmetic input for the GW-side. In current work with Lucy Yang, we show that over 2-adic local fields the resulting sequence has a duality symmetry.

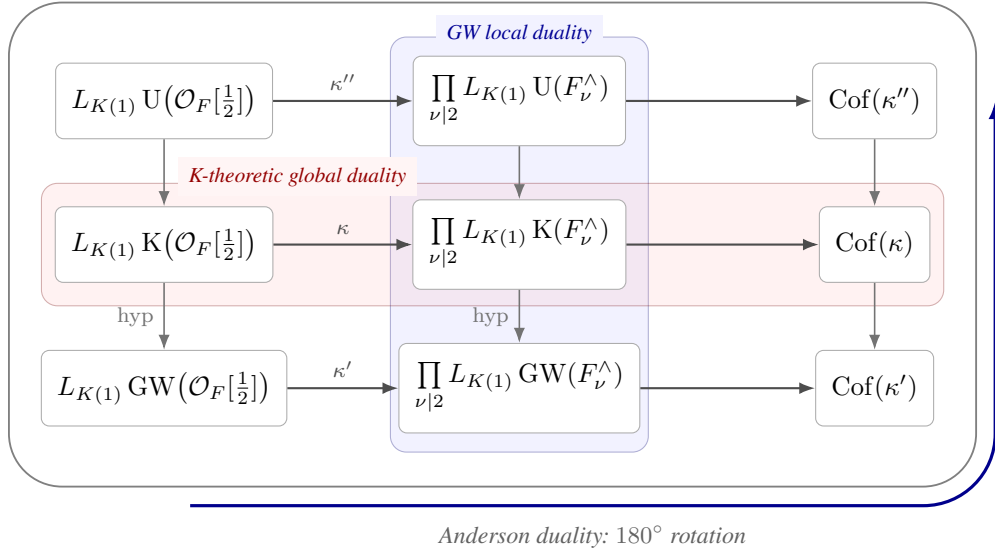
Theorem 3 ([CYon]). *Let k be a 2-adic local field. The hyperbolic sequence of $K(1)$ -local GW-theory admits a canonical 2-adic Anderson self-duality. The duality acts by reflection, interchanging the end terms and identifying the middle term with its own dual.*



The significance of this theorem is that the duality is compatible across the hyperbolic sequence $U \rightarrow K \rightarrow GW$ rather than appearing only in one theory at a time. It therefore suggests that the global duality should occur as one part of a larger self-dual arithmetic structure.

Our ongoing work seeks to globalize this duality to number fields.

Work in progress. *The completion sequences for U-theory, K-theory, and GW-theory fit into a (3×3) diagram compatible with the hyperbolic maps and local completion maps. The duality acts by half-turn rotation.*



Proving the self-duality of the full diagram would establish a global duality theorem for Grothendieck–Witt theory and combine the two results above in a single arithmetic structure. The rotational symmetry also guides the following conjecture.

Conjecture. *Let F be a number field with trivial involution, and let*

$$\kappa_{\text{KR}}^{[2]} : L_{K(1)} \text{KR}^{[2]}(\mathcal{O}_F[\tfrac{1}{2}]) \rightarrow \prod_{v|2} L_{K(1)} \text{KR}^{[2]}(F_v^\wedge)$$

denote the completion map, where the superscript $[2]$ refers to the shift of the duality on perfect modules. There is a canonical equivalence of genuine C_2 -spectra

$$I_{\mathbb{Z}_2}^{C_2} L_{K(1)} \mathrm{KR}(\mathcal{O}_F[\tfrac{1}{2}]) \simeq \mathrm{Cof}(\kappa_{\mathrm{KR}}^{[2]})$$

where $I_{\mathbb{Z}_2}^{C_2}$ denotes genuine C_2 -equivariant 2-adic Anderson duality.

After forgetting the C_2 -action, the shift of the duality disappears, and the conjecture recovers the K -theoretic Tate–Poitou duality of my dissertation.

3 Real trace methods

The duality results above reveal structural relations in Real algebraic K -theory, but they do not by themselves provide methods for computation. Trace methods have long provided such tools in ordinary algebraic K -theory by approximating K -theory with more computable invariants such as topological Hochschild homology (THH) and topological cyclic homology (TC). In joint work with Teena Gerhardt, Liam Keenan, Juan C. Moreno, and J.D. Quigley, I develop *homological Real trace methods* that incorporate the additional equivariant structure present in the Real setting [CGK⁺26].

Ordinary THH carries a circle action, while Real topological Hochschild homology (THR) extends this to an $O(2)$ -action by incorporating a reflection symmetry. Trace methods in this setting must keep track of genuine equivariant operations, which we use to compute the continuous homology of TCR^- .



Figure 3: The $\mathbb{T}$ -action on THH extends in THR to an $O(2) \simeq \mathbb{T} \rtimes C_2$ -action.

A key feature of our method is its compatibility with equivariant multiplicative and power operations, and we use these to extract information about the continuous homology of TCR^- . Technically, this is realized by the following spectral sequence.

Theorem 4 ([CGK⁺26]). *For a C_2 -equivariant commutative ring spectrum A , there is a natural multiplicative spectral sequence*

$$H_*(\mathrm{THR}(A); \mathbb{F}_2) \langle \bar{y} \rangle \Longrightarrow H_*^c(\mathrm{TCR}^-(A); \mathbb{F}_2),$$

compatible with equivariant power operations. For $MU_{\mathbb{R}}$, $BP_{\mathbb{R}}$, and several truncated Brown–Peterson theories, we determine the differentials and the E_{∞} -page of the resulting spectral sequences.

Our construction is a Real refinement of the homological trace methods of Bruner and Rognes [BR05], which played an important role in the study of Segal conjectures for the THH of bordism theories, MU and BP . A natural next test for our methods is a Real analogue of this for $MU_{\mathbb{R}}$ and $BP_{\mathbb{R}}$. Proving a Real Segal conjecture in these cases would extend the present calculations toward a broader understanding of Real trace invariants and ultimately of Real algebraic K -theory of Real bordism theories.

The way equivariant operations enter these computations makes their algebraic structure a problem in its own right. Beyond individual calculations, one must understand how the additional algebraic structures of equivariance fit together. The next section presents two projects concerned with different aspects of this structure.

4 Equivariant algebra: operations and gradings

A equivariant theory of descent must account for algebraic structure that disappears after forgetting the group action. Two features are particularly important: the equivariant operations carried by the coefficients and the representation gradings on which these operations act. My work investigates these two aspects of equivariant algebra through transfer systems and representation-graded Mackey and Tambara functors. These projects also clarify the kind of coefficient structure that a future equivariant descent theory should accommodate.

4.1 Equivariant operations. In equivariant stable homotopy theory, additive and multiplicative structures take forms that do not occur in ordinary algebra. Transfer maps play the role of equivariant addition, while norm maps play the role of equivariant multiplication. Each of these operations is parametrized by an N_∞ -operad, and a pairing of the two operads encodes an equivariant distributive law. Transfer systems describe N_∞ -operads combinatorially, and the distributive law becomes a compatibility condition between two such systems. The realization problem asks whether every compatible pair of transfer systems arises from a pairing of N_∞ -operads.

In joint work with David Chan, David Mehrle, Pablo Sanchez Ocal, Angélica Osorno, Ben Szczesny, and Paula Verdugo, we study the realization problem relating these two descriptions.

$$\begin{array}{ccc} \text{pairing of} & \xrightarrow{\quad\quad\quad} & \text{compatible pair of} \\ N_\infty\text{-operads} & \xleftarrow[\text{partially}]{} & \text{transfer systems} \end{array}$$

Theorem 5 ([CCM⁺25]). *A pairing of N_∞ -operads determines a compatible pair of transfer systems. Conversely, if the additive transfer system is complete, then the compatible pair can be realized by a pairing of N_∞ -operads.*

Beyond the complete case, we establish several additional families of realizable pairs and conjecture that every compatible pair of transfer systems is realizable.

4.2 Representation gradings. The next question concerns the grading of the objects on which these operations act. For a C_2 -spectrum X , its homotopy groups $\pi_* X$ form Mackey functors graded by representations of C_2 . An ordinary $RO(C_2)$ -grading records only the class of a representation and not its symmetries. In the category $\mathcal{RO}(G)$ introduced by Angeltveit and Bohmann in [AB18], morphisms between representation degrees also record the symmetries of individual representations.

$$\begin{array}{ccc} \text{\textit{RO}(C_2)\text{-grading}} & & \text{\textit{RO}(C_2)\text{-grading}} \\ \begin{array}{c} \pi_{[V]}^{C_2} X \\ \text{res} \downarrow \uparrow \text{tr} \\ \pi_{|[V]|}^e X \end{array} & & \begin{array}{c} \pi_V^{C_2} X \\ \text{res} \downarrow \uparrow \text{tr} \\ \pi_{|V|}^e X \end{array} \quad \begin{array}{c} \curvearrowright \\ \text{Aut}(V) \end{array} \end{array}$$

Figure 4: $\mathcal{RO}(C_2)$ -grading records the symmetries of the representation degree V .

In ongoing work with Noah Wisdom, we isolate the algebraic grading properties satisfied by the homotopy groups of C_2 -spectra. This defines a class of graded Mackey and Tambara functors that we call *topological*. Our main result identifies the additional information carried by the refined grading.

Theorem 6 ([CWon]). *The category of topological $\mathcal{RO}(C_2)$ -graded Mackey functors is equivalent to the category of topological $RO(C_2)$ -graded Mackey functors equipped with compatible actions of the grading automorphism groups. The same equivalence holds for topological Tambara functors.*

Thus, the additional information in a topological $\mathcal{RO}(C_2)$ -grading is precisely the symmetry data attached to the ordinary $RO(C_2)$ -graded homotopy groups.

5 Outlook

A main direction of my research is to understand duality and descent in Real algebraic K -theory. The conjecture above gives a concrete target for this direction. My dissertation derives K -theoretic Tate–Poitou duality from étale duality through Thomason descent. For KR , no corresponding representation-graded equivariant cohomology theory is yet available, so I approach the problem from the other direction. The conjecture specifies the duality that a Real descent theory should produce. Such a theory must recover ordinary Thomason descent after forgetting the C_2 -action and be compatible with the expected global Grothendieck–Witt duality in the 3×3 diagram. I aim to construct such a cohomology theory together with a descent spectral sequence converging to KR .

My Real trace project studies KR through explicit calculations of its trace invariants. Calculations of the continuous homology of TCR^- for $MU_{\mathbb{R}}$, $BP_{\mathbb{R}}$ are the starting point for a Real Segal conjecture for $MU_{\mathbb{R}}$ and $BP_{\mathbb{R}}$. Proving this conjecture would provide computable approximations to $KR(MU_{\mathbb{R}})$ and $KR(BP_{\mathbb{R}})$.

My work on transfer systems and representation gradings studies the algebra specific to genuine equivariance. These projects describe how operations coexist and how symmetries act across representation degrees. The same structures will occur in the coefficients of a future equivariant descent theory.

The program therefore ranges from number rings to equivariant ring spectra. My broader aim is to develop structural and computational methods for Real algebraic K -theory of rings and ring spectra with involution.

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